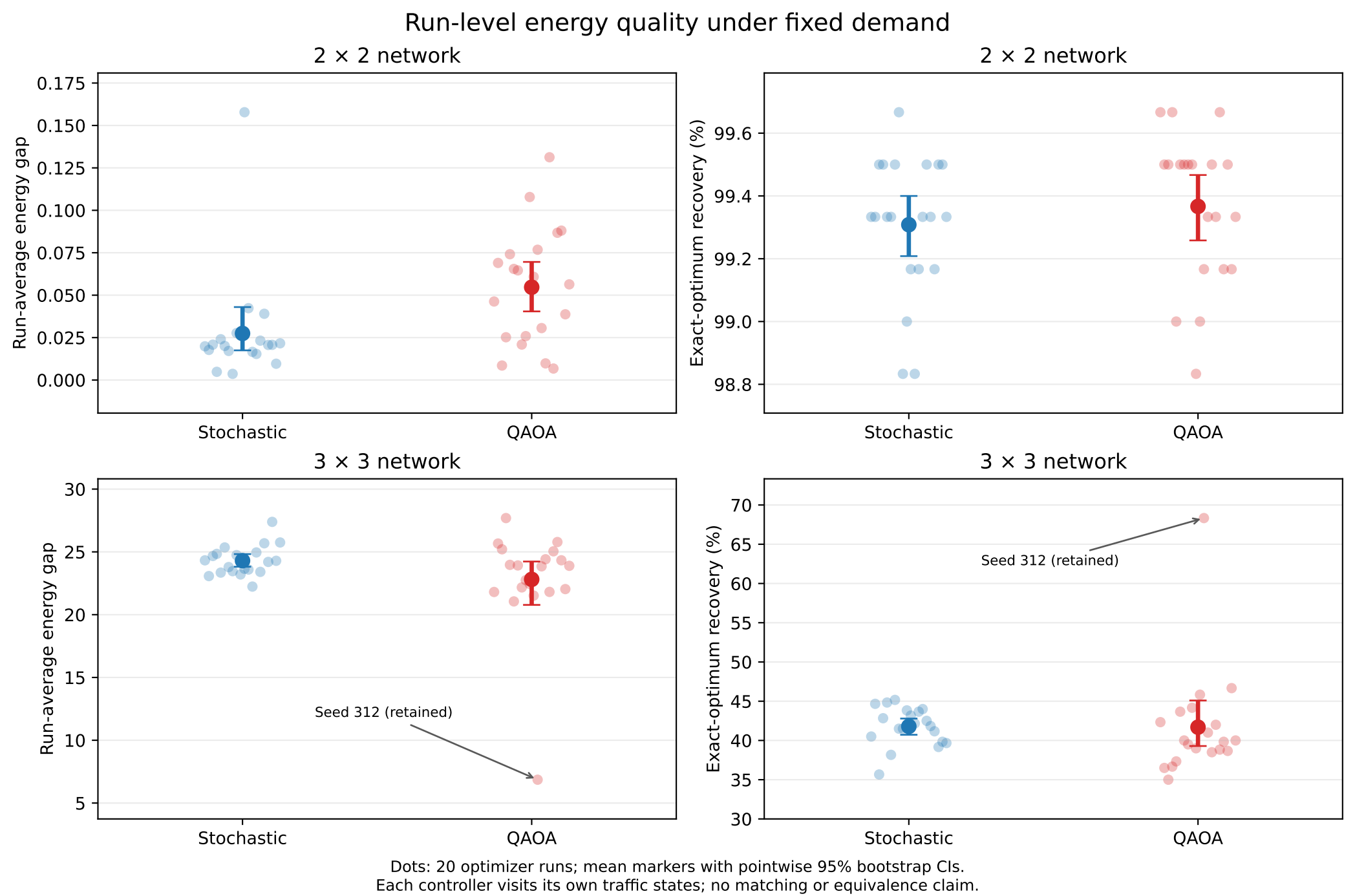
Run-level energy quality under fixed demand
2 × 2 network
Run-average energy gap
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Stochastic
QAOA
2 × 2 network
Exact-optimum recovery (%)
99.6
99.4
99.2
99.0
98.8
Stochastic
QAOA
3 × 3 network
Run-average energy gap
30
25
20
15
10
5
Seed 312 (retained)
Stochastic
QAOA
3 × 3 network
Exact-optimum recovery (%)
70
65
60
55
50
45
40
35
30
Seed 312 (retained)
Stochastic
QAOA
Dots: 20 optimizer runs; mean markers with pointwise 95% bootstrap CIs.
Each controller visits its own traffic states; no matching or equivalence claim.